

Effect of high alkalinity on shrimp gills: Histopathological alternations and cell specific responses

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ABSTRACT

High alkalinity stress was considered as a major risk factor for aquatic animals surviving in saline-alkaline water. However, few information exists on the effects of alkalinity stress in crustacean species. As the dominant role of gills in osmotic and ionic regulation, the present study firstly evaluated the effect of alkalinity stress in *Exopalaemon carinicauda* to determine changes in gill microstructure, and then explore the heterogeneity response of gill cells in alkalinity adaptation by single-cell RNA sequencing (scRNA-seq). Hemolymph osmolality and pH were increased remarkably, and gills showed pillar cells with more symmetrical arrangement and longer lateral flanges and nephrocytes with larger vacuoles in high alkalinity. ScRNA-seq results showed that alkalinity stress reduced the proportion of pillar cells and increased the proportion of nephrocytes significantly. The differentially expressed genes (DEGs) related to ion transport, especially acid-base regulation, such as $V(H^+)$ -ATPases and carbonic anhydrases, were down-regulated in pillar cells and up-regulated in nephrocytes. Furthermore, pseudotime analysis showed that some nephrocytes transformed to perform ion transport function in alkalinity adaption. Notedly, the positive signals of carbonic anhydrase were obviously observed in the nephrocytes after alkalinity stress. These results indicated that the alkalinity stress inhibited the ion transport function of pillar cells, but induced the active role of nephrocytes in alkalinity adaptation. Collectively, our results provided the new insight into the cellular and molecular mechanism behind the adverse effects of saline-alkaline water and the saline-alkaline adaption mechanism in crustaceans.

1. Introduction

Soil salinization and alkalization are major environmental problems formed by long-term evaporation, climate drought, or little rain threats to the agriculture industry worldwide and affect approximately 20% of the world's cultivated land and nearly half of all irrigated land, and the impact is becoming increasingly severe (Zhu, 2001). A lot of saline-alkaline water are distributed throughout these areas, which are characterized by high alkalinity, high pH, poor buffering capacity and various types of ion imbalances (Lin et al., 2013). Saline-alkaline water will substantially impact the survival, development, and reproduction of organisms, resulting in decreased biomass and considerable changes in species composition (Liu et al., 2022). High alkalinity stress was considered as a major risk factor for aquatic animals, which can pose several physiological challenges, even cause degradation of cellular

biomolecules (Yao et al., 2015). A typical case is that high alkalinity is always combined with high pH and leads to the decrease of H^+ concentration in fish. Thus the toxic ammonia nitrogen (NH_3) cannot be combined with H^+ and cannot be discharged in the form of non-toxic ammonium (NH_4^+) in vivo (Zhao et al., 2020). Therefore, high alkalinity often negatively affects nitrogen waste excretion for fish and induce metabolic alkalosis. In addition, some aquatic animal exposure to high alkaline pH can prevent Na^+ influx, causing an imbalance of osmotic pressure inside and outside the body, inhibiting growth and even causing death (Song et al., 2021). Consequently, high alkalinity conditions may significantly affect ion transport and osmotic regulation for aquatic animals.

Gills act as a direct interface between hemolymph or blood of aquatic animals and the ambient environment, which are generally multifunctional, performing the majority of an animal's respiratory gas exchange,

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osmoregulation, acid-base balance, and excretion of nitrogenous end products while also acting as a major toxicological interface for metals and pollutants (Allen and Weihrauch, 2021; McNamara and Faria, 2012). Therefore, gills play a key role in response to saline-alkaline stress in aquatic animals. At present, the research on gill cell classification of aquatic animals mainly focuses on fish. In fish, the gill tissue mainly contains the ionocytes, mucous cells, pavement cells, pillar cells, blood cells. The ionocytes are the main sites of ion transports in gills. In *Oreochromis alcalicus graham*, the ionocytes of gills showed the characteristics of seawater type ionocytes with honeycomb surface and less microvilli under extreme alkalinity condition (Maina, 1990). The pavement cells of *Leuciscus waleckii* inhabiting Dali Nor lake were proliferating, which may prevent HCO_3^- entering blood in order to adapt the high alkalinity environment (Gao et al., 2020). The number of mucous cells in *Chalcalburnus tarichi* was significantly decreased during migration from alkaline water to freshwater (Oğuz, 2015). Therefore, the various gill cells in aquatic animals changed significantly in response to alkalinity stress. Although given the importance of gills, many omics methods such as transcriptome and proteome have been applied to study their responses after alkalinity stress at the ensemble, cell population levels, these studies did not focus on the heterogeneity of different gill cell types (Chen et al., 2020; Su et al., 2020). Recently, the single-cell RNA sequencing (scRNA-seq) techniques designed to reveal the toxicological mechanisms at a single-cell resolution could help to illustrate the consequence of organism complexity and cell heterogeneity (Meng and Wang, 2022).

Although there are a lot of research on the physiological and molecular mechanism behind the adverse effects of saline-alkaline water on fish and the regulatory mechanism in fish tolerance, for crustaceans, another important aquatic animal, their responses to alkalinity stress remain unknown. To date, although there are some researches on the alkalinity tolerance range and physiological change of crustaceans, the researches on the deeply cellular and molecular mechanism in response to saline-alkaline stress were barely (Qin et al., 2021). The ridgetail white prawn, *Exopalaemon carinicauda*, has great adaptive capacity to extreme environments, especially for salinity, pH and carbonate alkalinity (Shi et al., 2022). In recent years, *E. carinicauda* has successfully cultivated in saline-alkaline waters in China, which make it attractive as an excellent model for studying the response mechanism under alkalinity stress in crustacean (Ge et al., 2019). Although gills play an important role in maintaining ion-base balance in the body of crustaceans, the information on the gill cell classification and the cell characteristics in crustaceans is less compared with fishes, which limit to understand the deeply cellular and molecular mechanism in response to alkalinity stress.

Therefore, the aims of this study were to test the effects of alkalinity stress in gills of *E. carinicauda* in order to understand their resistance or vulnerability to saline-alkaline challenge. More specifically, we analyzed the effects of alkalinity stress on physiological responses and histological structure of gills in *E. carinicauda*. In addition, we performed single-cell RNA sequencing (scRNA-seq) analyses of gills in response to alkalinity stress, focusing on cell classification and heterogeneity. This study will provide comprehensive resources for better understanding the cellular and molecular mechanism behind the adverse effects of saline-alkaline water on crustacean and the regulatory mechanism in crustacean tolerance.

2. Materials and methods

2.1. Alkalinity stress experiment

Healthy adult *E. carinicauda* (body length 3.5 ± 0.3 cm, body weight 1.25 ± 0.08 g) were obtained from a commercial farm in Rizhao, China and acclimated in 200 L polyvinyl chloride polymer (PVC) tanks containing aerated and sand-filtrated seawater (salinity 31 ppt, pH 8.0 $\pm$ 0.1, carbonate alkalinity 2.5 ± 0.5 mmol/L) at 23 ± 1 °C for 2 weeks.

After acclimation period, shrimps were transferred to six tanks containing 60 shrimps per tank. Three tanks were considered as the control group, and the other three tanks were exposed to carbonate alkalinity at 10 mmol/L according to previous study, which were considered as the alkalinity group (Ge et al., 2019). The alkalinity was adjusted using NaHCO_3 and measured by an acid-based titration method (Ge et al., 2019). Hemolymph from six shrimps in each tank were sampled to analyze plasma osmolality and pH at high alkalinity exposure times of 12 h, 24 h, 48 h, 72 h and 168 h with 0 h as natural alkalinity control group. At 48 h, gills from three shrimp in each tank were sampled for the histological evaluation, electron microscopy observation and frozen section, and gills from three shrimps in each tank were collected and mixed as one alkalinity sample for scRNA-seq.

2.2. Hemolymph osmolality and pH measurements

Hemolymph pH was analyzed immediately after sampling, using a pH Tester (HI98103, HANNA, Italy). The remaining hemolymph was centrifuged at 4000 g for 20 min at 4 °C (Eppendorf centrifuge 5810 R) to obtain serum. The serum was analyzed for osmolality. Serum osmolality was measured in undiluted 70 μL aliquots using a freezing point osmometer (SMC 30 C-1, Tianhe, China).

2.3. Histological and ultrastructural analyses of gills

Histological assay: Gills were fixed in Bonn's fixative for 24 h, and then transferred to 70% ethyl alcohol for storage. Subsequently, gills were dehydrated with a graded ethanol series and embedded in paraffin wax. After dehydration and embedding, the fixed tissues were sectioned (3 μm) in a microtome. Then, the sections were stained with haematoxylin and eosin (H&E) and examined under a microscope (Guo et al., 2021).

Scanning electron microscopy (SEM): Gills were fixed in 2.5% glutaraldehyde in 0.1 M PBS, pH 7.4, at 4 °C for 24 h and were post-fixed in 1% Osmium Tetraoxide (OsO_4) in 0.1 M PBS, pH 7.4 for 1 h and then dehydrated with a graded increasing ethanol series. Tissues were coated with platinum and palladium in a Hitachi ion-sputtering apparatus (E2500). The processed tissues were examined in a Hitachi S-2500 scanning electron microscope operating at 15 KV (Arumugam et al., 2021).

Transmission electron microscopy (TEM): Gills were fixed in 2.5% glutaraldehyde in 0.1 M PBS, pH 7.4, at 4 °C for 24 h and were post-fixed by 1% OsO_4 . After dehydration in ascending graded alcohol and clearness by propylene oxide, tissues were molded with epoxy resin, kept in an incubator at 60 °C for 48 h, cooled down, sectioned (1 μm) through ultra-microtome (Leica ultracut UCT) with a glass knife, and stained by toluidine blue (Arumugam et al., 2021).

2.4. Single-cell isolation of gill

The fresh gills in control and alkalinity groups were immediately used for their single cell suspension. Briefly, after being washed 2–3 times with DPBS (Na_2HPO_4 8 mM, NaCl 136 mM, KH_2PO_4 2 mM, pH 7.8) and cut into pieces, the gills were successively digested with collagenase (2 mg/mL) and disperse enzyme (0.2 mg/mL) at 37 °C for 15 min. The digested solution was centrifuged to collect the cell sediment and the pre-cooled CMFSS (calcium free magnesium ion buffer, 25.5 g/L NaCl, 0.8 g/L Na_2SO_4 , 2.86 g/L HEPES) was used to separate gill cells in the cell sediment (Sun et al., 2021). Subsequently, the cell suspension was filtered through the 40 μm cell strainer to remove large particles, and then centrifuged at 500 g for 10 min. Cell viability and number were assessed by 0.4% trypan blue staining and cell counting using a cell counter. High-quality cell suspension of gill cells was subjected to single cell encapsulation by 10X Genomics v3 kit (10x Genomics, USA).

2.5. Single cell RNA-sequencing

Single cell encapsulation, complementary DNA (cDNA) library synthesis, RNA-sequencing and data analysis were completed by Gene Denovo company (Guangzhou, China). In brief, the cell suspension was adjusted to ensure the cell density was greater than 1000 cells/ μ L. Cells were barcode-labeled and mixed with reverse transcriptase into single-cell Gel Beads-In-Emulsions (GEMs), then the cDNA library was constructed by reverse transcription and global amplification in a Mastercycler Pro thermocycler (Eppendorf, Hamburg, Germany). The cDNA library was sequenced on the Illumina novaseq6000 platform (Sun et al., 2021). The BioProject number submitted to NCBI is PRJNA882579.

2.6. Data processing and analysis

Due to the genome sequence of *E. carinicauda* is still unknown, the full-length transcriptome data was prepared as a reference for mapping scRNA-Seq data. The raw data were aligned, filtered, and normalized using 10X Genomics Cell Ranger software (version 3.1.0). Briefly, reads with low-quality barcodes and Unique Molecular Identifiers (UMIs) were filtered out and after the initial quality control screen by the CellRanger, the sequencing data were aligned to the reference transcriptome. Subsequently, the cell by gene matrices for each sample were individually imported to a popular R package Seurat for data normalization, cell clustering, marker gene screening, and gene expression analysis among different cell clusters. To start, cells with unusually high number of UMIs (≥ 8000) or mitochondrial gene percent ($\geq 10\%$) were filtered out and excluded cells with less than 500 or more than 4000 genes detected. After removing of low-quality cells and normalizing data were normalized with the formula: Gene expression = $\log(1 + (\text{UMI} \div \text{UMITotal}) \times 10,000)$. Then, the normalized data were clustered using principal component analysis (PCA) and visualized with t-distributed Stochastic Neighbor Embedding (t-SNE). The cell subset was grouped by graph-based clustering based on the gene expression profile of each cell. Expression value of each gene in given cluster were compared against the rest of cells using Wilcoxon rank sum test. Differently expressed genes (DEGs) in each cluster were identified by the following criteria: (1) $p\text{-value} \leq 0.01$; (2) $\log_2\text{FC} \geq 0.360674$ (FC is the fold-change in the average expression between the two groups); and (3) genes expressed in more than 25% of the cells. The top 5 up-regulated genes in each cluster were used for plotting maps. DEGs were also identified after alkalinity stress. A hurdle model in the model-based analysis of scRNA-seq was used to identify the DEGs between the control and alkalinity group in one cluster with the following criteria: (1) $|\log_2\text{FC}| \geq 0.585$; (2) $p\text{-value} \leq 0.05$; and (3) the percentage of cells in which the gene was detected in a specific cluster $> 25\%$ (Meng and Wang, 2022). To investigate potential gene functions and pathways in cell clusters, the up-regulated genes in each cluster were further analyzed for GO and KEGG enrichment (Zhu et al., 2022).

2.7. Pseudotime analysis

Monocle was used to constructing single-cell trajectories in pseudotime according to differentially expressed genes. We assumed the raw UMI counts were distributed according to a negative binomial distribution with fixed variance expression family to model the raw UMI count data, as recommended by the authors of Monocle (Qiu et al., 2017). The Monocle function BEAM was used to identify cells that were enriched along particular branches in the pseudotime tree. Branched heat maps were constructed using genes with false discovery rate (FDR) $< 5 \times 10^{-5}$ from the BEAM.

2.8. Fluorescent in situ hybridization (FISH)

The fluorescein-labeled probes of some genes were synthesized using the fluorescein RNA Labeling Kit (Roche Diagnostics), including

cytoplasmic carbonic anhydrase (CAC), neuronal acetylcholine receptor subunit alpha-6 (NAR), prophenoloxidase (ProPO), Na^+/K^+ -ATPase alpha (NKA- α), vitellogenin (Vg) and peroxidase, which was selected as gene marker of cluster 0–5, respectively. The riboprobe sequences are provided in Table S1. In order to perform polychromatic in situ hybridization, the digoxigenin (DIG)-labeled riboprobes were synthesized from the coding sequence of the NAR gene using the DIG RNA Labeling Kit (Roche Diagnostics, Mannheim, Germany). Frozen sections of gills were generated using a Leica CM1950 microtome. Tissue sections were first prehybridized for 20 min, and then hybridization buffer containing DIG-labeled or fluorescein-labeled riboprobes were added to each slide. After hybridization with 70°C for 12 h, sections were washed thrice in wash solution at 70°C for 30 min, and TNT (1 M Tris-HCl, 5 M NaCl, 0.1% Tween-20, pH 7.5) at room temperature for 10 min, then added in Glycine-HCl for 10 min and then washed thrice in TNT at room temperature for 10 min. Next, the slides were blocked with blocking fluid at 23°C for 3 h and the primary anti-Fluorescein-POD (Thermo fisher) was added and incubated overnight at 4°C . After rinse with TNT, the slides were incubated with Cy3 conjugated second antibody. If continue to do polychromatic FISH, the primary anti-DIG-peroxidase (Thermo fisher) was added and incubated with Cy5 conjugated second antibody. Finally, DAPI (Sigma-Aldrich, St. Louis, MO, USA) was further used for staining cell nuclei in the slides. Fluorescent signals from FISH were imaged using a Zeiss confocal microscope (Oberkochen, Germany).

2.9. Statistical analysis

For hemolymph osmolality and pH analysis, data were analyzed with one-way ANOVA followed by Fisher's least significant difference (LSD) test by using SPSS 19.0 software (Chicago, IL). The student's t-test was subjected to determine the significant differences among different sampling time points. All the presented data showed the mean $\pm$ standard deviation (SD) from all the replicates.

3. Results

3.1. Hemolymph osmolality and pH

After alkalinity stress, the osmolality of hemolymph increased first, followed by recovery to control level, and the maximum value appeared at 48 h. Hemolymph pH showed a steady increase for the first 48 h, and then dropped a bit, but it was still higher than control level at 168 h after alkalinity stress (Table 1).

3.2. Gill histological structure

The gill of *E. carinicauda* is typical phyllobranchiate including 7 pairs of gill filaments, which composed of a central axis and one row of flattened lamellae on each side. SEM images showed relatively smooth gill filament surfaces in control group, and became wrinkled and rough with some small pores appeared in the central gill axis after alkalinity stress (Fig. 1A, B and b). The histological analysis showed that nephrocytes were observed only in the central axis with a voluminous vacuole (Fig. 1C). After alkalinity stress, the volume of vacuole in nephrocytes increased (Fig. 1D). Compared with the control group, the TEM results of gill lamellar showed that the lateral pillar cell flanges obviously lengthened, the pillar cells showed more symmetrical arrangement, and the hemolymph space decreased after alkalinity stress (Fig. 1E and F). In addition, the positive signals of the CAC (marker gene of pillar cell) were detected by FISH both in gills of control and alkalinity group. The CAC-positive signal of pillar cells became more symmetrically distributed after alkalinity stress, which was in line with the result of TEM (Fig. 1G and H). It follows that the structure of pillar cell and nephrocyte changed obviously after alkalinity stress, which indicated that these two cell types may play important roles in response to alkalinity stress in *E. carinicauda*.

Table 1
Hemolymph osmolality and pH in *E. carinicauda* under alkalinity stress.

Time (h)	0	12	24	48	72	168
Osmolality (mOsm/kg)	760.67 ± 8.02 ^a	790.33 ± 7.09 ^b	818.33 ± 4.73 ^c	826.33 ± 4.16 ^c	784.67 ± 25.54 ^{ab}	765.00 ± 15.13 ^a
pH	7.65 ± 0.03 ^a	7.8 ± 0.05 ^b	7.93 ± 0.02 ^c	8.19 ± 0.03 ^e	8.15 ± 0.04 ^{de}	8.08 ± 0.05 ^d

Different superscript letters in each line indicate significant differences ($P < 0.05$).

3.3. ScRNA-seq clustering of gill

ScRNA-seq was used to classify the cell types and investigate the heterogeneity responses of gill cells before and after alkalinity stress (Fig. 2A). After standard quality control dataset filters, eight populations (cluster 0–7) were obtained that can be visualized using t-SNE (Fig. 2B). A total of 10127 cells (control: 4731, alkalinity: 5396) from cell suspensions of whole gill cells were isolated and sequenced. These cell clusters consisted of 0.3–49.38% cells, and cluster 0 accounted for nearly half (Fig. 2C). The expression levels and the percentage of cells expressing the top five genes in each cluster are shown in a dot plot (Fig. 2D). These genes need to be confirmed in future research and are proposed to be used as marker genes for each cluster of *E. carinicauda* gill cells.

Up to now, no enough known marker genes exist that can be used to specify different cell types of gills in crustacean. To characterize the clusters, we identified DEGs and inferred putative identities based on only a few known markers. Most typical ion transport related genes, such as V-H⁺-ATPase (VHA) subunits (Lucena et al., 2015), CAs (Huang et al., 2019), Rh (Egnew et al., 2019) and NCC (Barany et al., 2021), were widely expressed in the cluster 0. In view of VHA and CA were mainly localized in the pillar cells of *M. amazonicum* (Maraschi et al., 2015), the cluster 0 may be identified as pillar cells. In addition, a small number of ion transport related genes were present in cluster 3, such as potassium voltage-gated channel, glutamate-gated chloride channel 3, NKA- α and NKA- β subunit. In *M. amazonicum*, NKA labeling was restricted to the septal cells of gill, thus we preliminarily identified cluster 3 as septal cells. Some typical marker genes of hemocytes, such as prophenoloxidase (Teresa et al., 1999), integrin (Koiwai et al., 2018), interleukin (Zhao et al., 2021) and crustin (Afsal et al., 2013), were widely expressed in cluster 2, thus cluster 2 may be named hemocytes (Fig. S1A). As for other clusters, no known marker genes exist that can be used to specify their cell types. GO enrichment analysis of the DEGs in each cluster were performed to identify their cell types (Fig. S1B). Cluster 0 was mainly related to ion transport, Cluster 2 was mainly related to response to stimulus and immune system, Cluster 3 was mainly enriched in “electron transport chain”, which were all consistent with the function corresponding their cell types. Cluster 1 was mainly enriched in “protein targeting to membrane” and “co-translational protein targeting to membrane”, but the cell type cannot be determined from this result. Cluster 4 was mainly enriched in “neuron differentiation”, “generation of neurons and neuron development”, which may be identified as nerve cell. Cluster 5 was mainly enriched in “uronic acid metabolic process”, “glucuronate metabolic process and RNA processing”, which may be identified as nurse cell. Besides the above six clusters, cluster 6 and cluster 7 with only several cells were identified, but is meaningless for further analysis. KEGG enrichment analysis of the DEGs in each cluster were also performed, and “Oxidative phosphorylation” pathway was significantly enriched in both cluster 0 and cluster 3 (Fig. S2).

The marker genes of each cluster were selected to further determine the localization of each cell type in gill (Fig. 3A). Pillar cell, nephrocyte, hemocyte, septal cell, nerve cell and nurse cell were marked by CAC, NAR, ProPO, NKA- α , Vg and Peroxidase gene, respectively. The expression of the marker gene in each cluster was confirmed by FISH technology. The positive signals of CAC, ProPO, NKA- α , Vg and Peroxidase were mainly localized in the gill lamellar, and the NAR was mainly localized in the central gill axis. Intense CAC-positive signal was

predominantly detected in the apical flanges of pillar cells. ProPO-positive signal was localized to the outer margin of hemocyte nuclei. The positive labeling of NKA- α marked the location of septal cell in gill lamellar. The labeling of Vg was not only detected in the outer margin of nerve cell nuclei, but also found in the inner nuclear layer of nuclei. Peroxidase-positive signal was mainly detected in the surrounding of nurse cell nuclei tightly. In view of the result that only NAR-positive signal was detected in the surrounding of vacuole in gill axis, which was according with the localization characteristics of nephrocytes, thus cluster 1 could be named as nephrocytes (Fig. 3B).

3.4. Heterogeneous responses of gill cells

The proportion of pillar cells decreased from 54.81% to 44.63%, while the nephrocytes increased from 12.56% to 26.39% significantly after alkalinity stress. The proportions of other clusters showed no significant change (Fig. 4A). A total of 1227 DEGs were identified between control and alkalinity stress group, and nephrocytes and pillar cells contained more DEGs than other cell types. In nephrocytes, the number of up-regulated DEGs was more than the down-regulated DEGs, while the result was opposite in pillar cell (Fig. 4B). The data revealed that pillar cells and nephrocytes changed significantly in cell number and molecular regulation after alkalinity stress. In the following, we mainly focused on their responsive mechanisms.

The Venn diagram showed that pillar cell and nephrocyte may had a similar response, because the number overlap of DEGs in a particular cell cluster reached up to 154 (Fig. 4C). The GO enrichment showed that DEGs were mainly assigned to hydron transport including “electron transport chain”, “hydrogen transport” and “proton transport” terms in both pillar cell and nephrocyte (Fig. 4D). Most of these DEGs showed significantly down-regulated expression in pillar cell and up-regulated expression in nephrocyte, which indicated that the nephrocytes may replace pillar cells to play a positive role in hydron transport in response to alkalinity stress (Fig. S3). The KEGG result showed that the most abundant pathway was “Oxidative phosphorylation” in both pillar cell and nephrocyte (Fig. 4E). Most DEGs in H⁺ and HCO₃⁻ transport related pathways including “Oxidative phosphorylation”, “Collecting duct acid secretion” and “Nitrogen metabolism” showed significantly down-regulated expression in pillar cell and up-regulated expression in nephrocyte after alkalinity stress, especially VHAs and CAs (Fig. S4). Above results revealed that the H⁺ and HCO₃⁻ transport function of pillar cell was inhibited, while that of nephrocyte was activated after alkalinity stress.

3.5. Pseudotime analysis of nephrocytes

In view of the positive role of nephrocyte in response to alkalinity stress, we selected it to perform the differentiation trajectory analysis. Monocle was used to study the single-cell transition process of nephrocyte cluster in both control and alkalinity group. In the unsupervised pseudotime analysis, nephrocyte differentiation was divided into three major tree-like structures. Two branches were found following the pre-branch, including branch 1 and branch 2, and the cell numbers in branch 2 increased significantly after alkalinity stress (Fig. 5A). GO enrichment analysis revealed that the branch 2 was significantly related to ion transport, including “electron transport”, “hydrogen transport”, “proton transport” and “monovalent inorganic cation transport” (Fig. 5B). The result of pseudotime analysis indicated that alkalinity stress

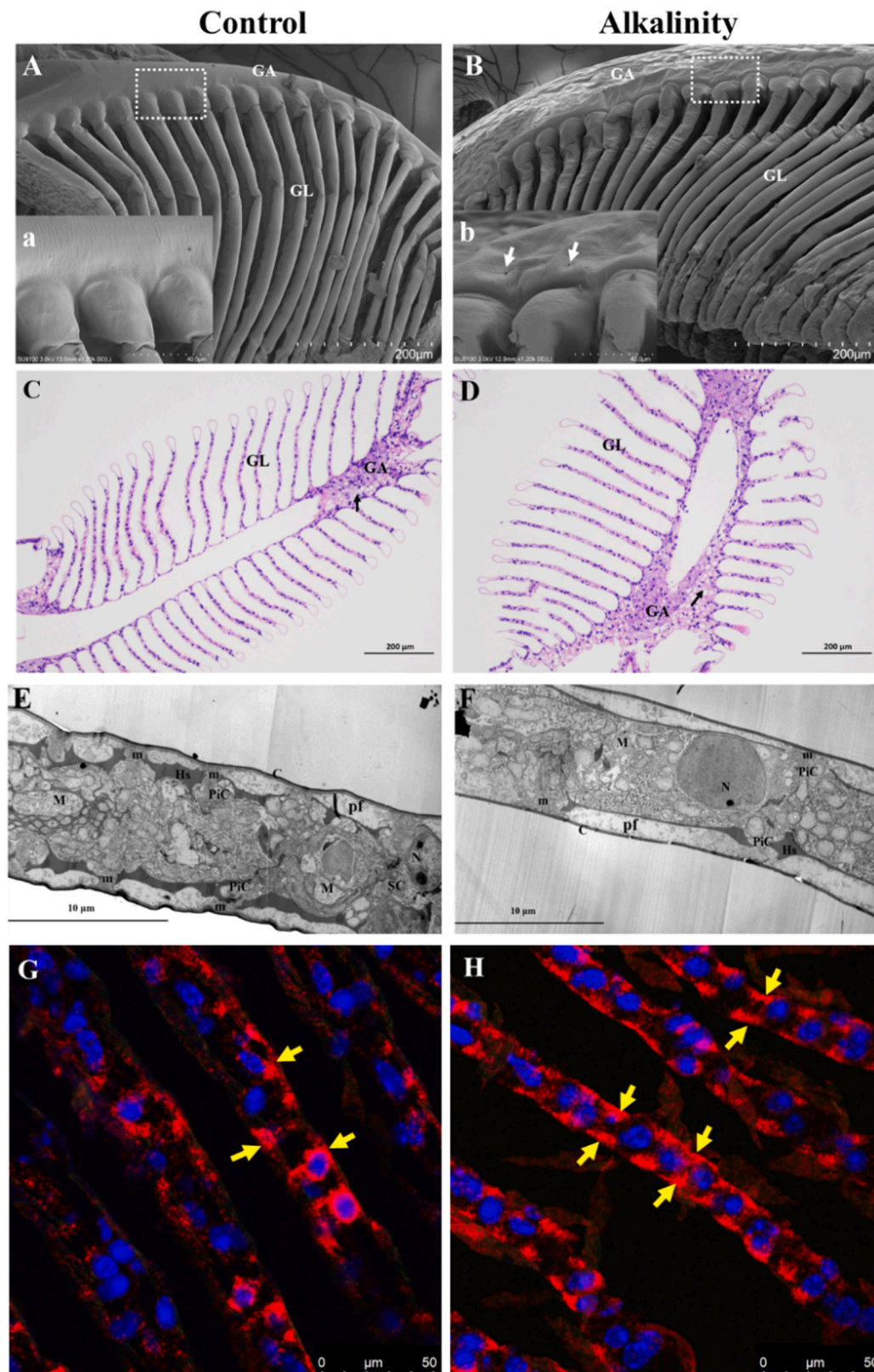


Fig. 1. Histological and cytological features of gill in *E. carinicauda*. (A) The gill filament surface in the control group; (a) enlarged view of the central axis in the control group. (B) The gill filament surface in the alkalinity group; (b) enlarged view of the central axis in the alkalinity group. (C) The gill histopathology in the control group. (D) The gill histopathology in the alkalinity group. (E) Ultrastructure of transverse sections of gill lamellae in the control group. (F) Ultrastructure of transverse sections of gill lamellae in the alkalinity group. GA gill axis; GL gill lamellar; c cuticle; Hs hemolymph space; M mitochondria; N nucleus; m microvillus; PiC pillar cells; SC septal cells; pf pillar cell flange. The white arrow pointed to pore. The black arrow pointed to the vacuole in nephrocytes (NC). (G) The CAC location in the pillar cells of control group. (H) The CAC location in the pillar cells of alkalinity group. The yellow arrow pointed to the CAC-positive signal.

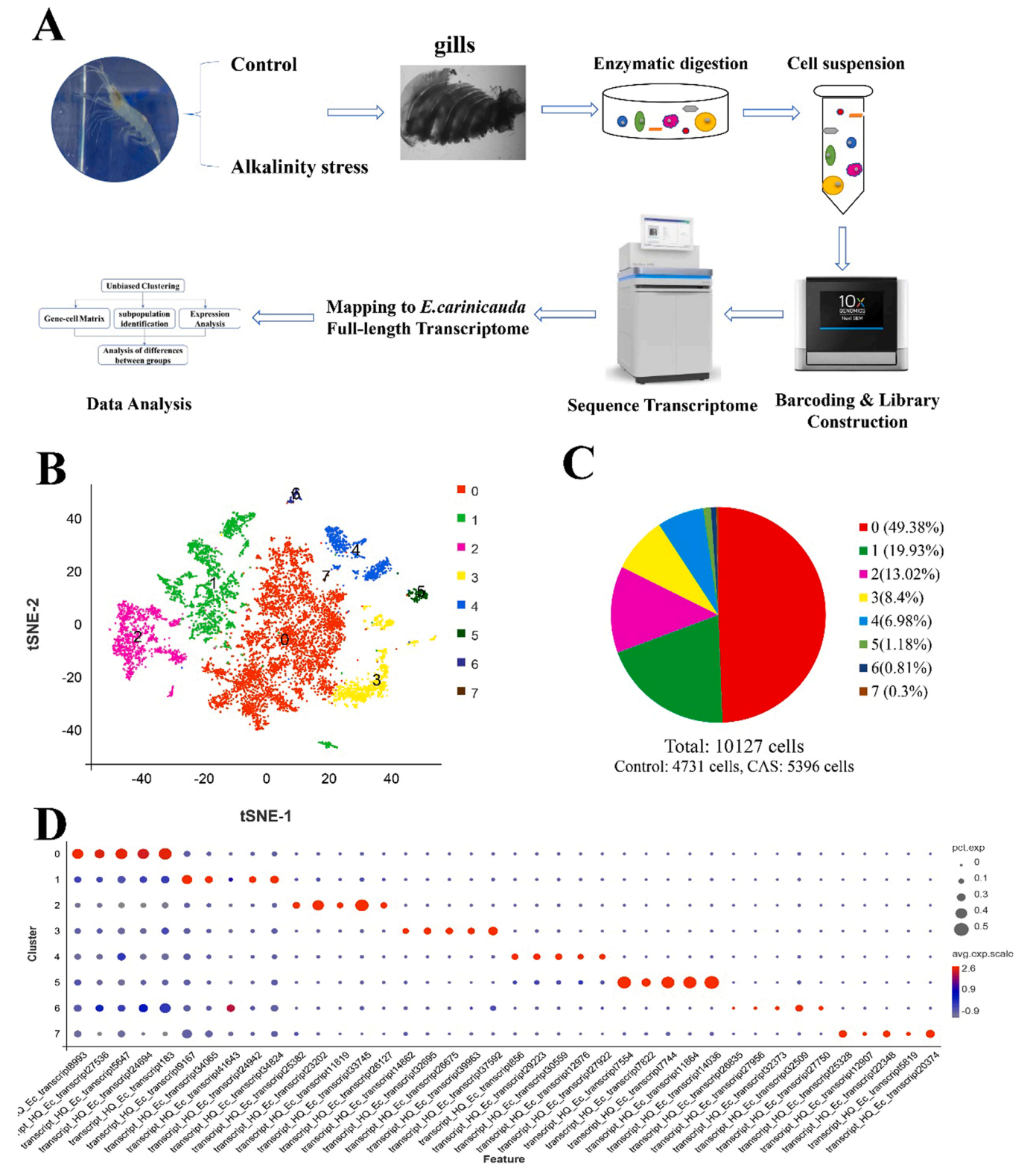


Fig. 2. scRNA-seq analysis of gill cells in *E. carinicauda*. (A) Overall workflow for cell sorting and single-cell data analyses. (B) The t-SNE nonlinear clustering of gill cells in *E. carinicauda*. (C) Number of cells in each cluster and their proportional distribution in the total gill dataset. (D) Top 5 DEGs (x axis) identified in each cluster (y axis). Dot size represents the fraction of cells in the cluster that express the gene; intensity indicates the mean expression (Z-score) in expressing cells, relative to other clusters.

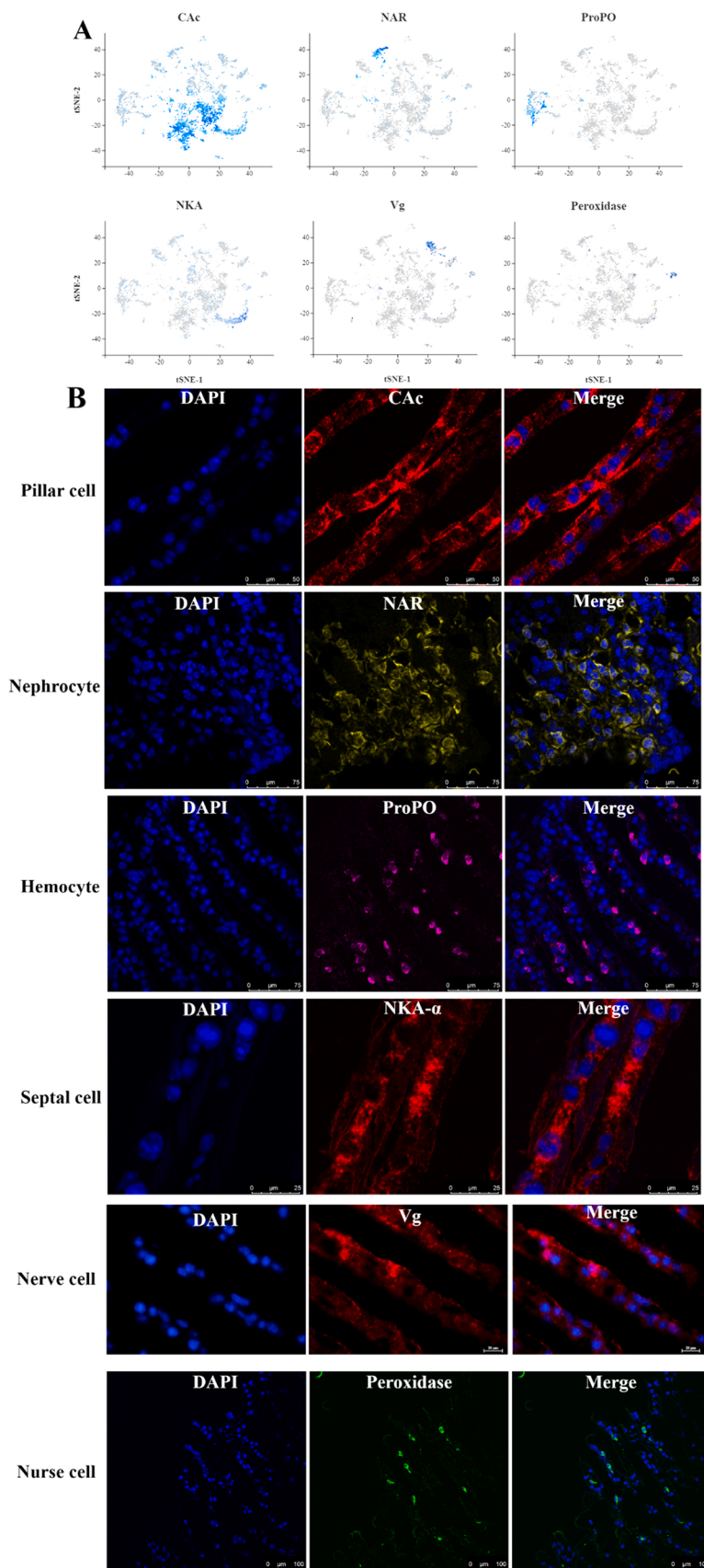


Fig. 3. Overview of the gill cell clusters in *E. carinicauda*. (A) t-SNE plots showing expression of marker genes in cluster 0–5. The gene expression level is color coded. (B) FISH image showing the location of marker genes in different cell types in the gill filament.

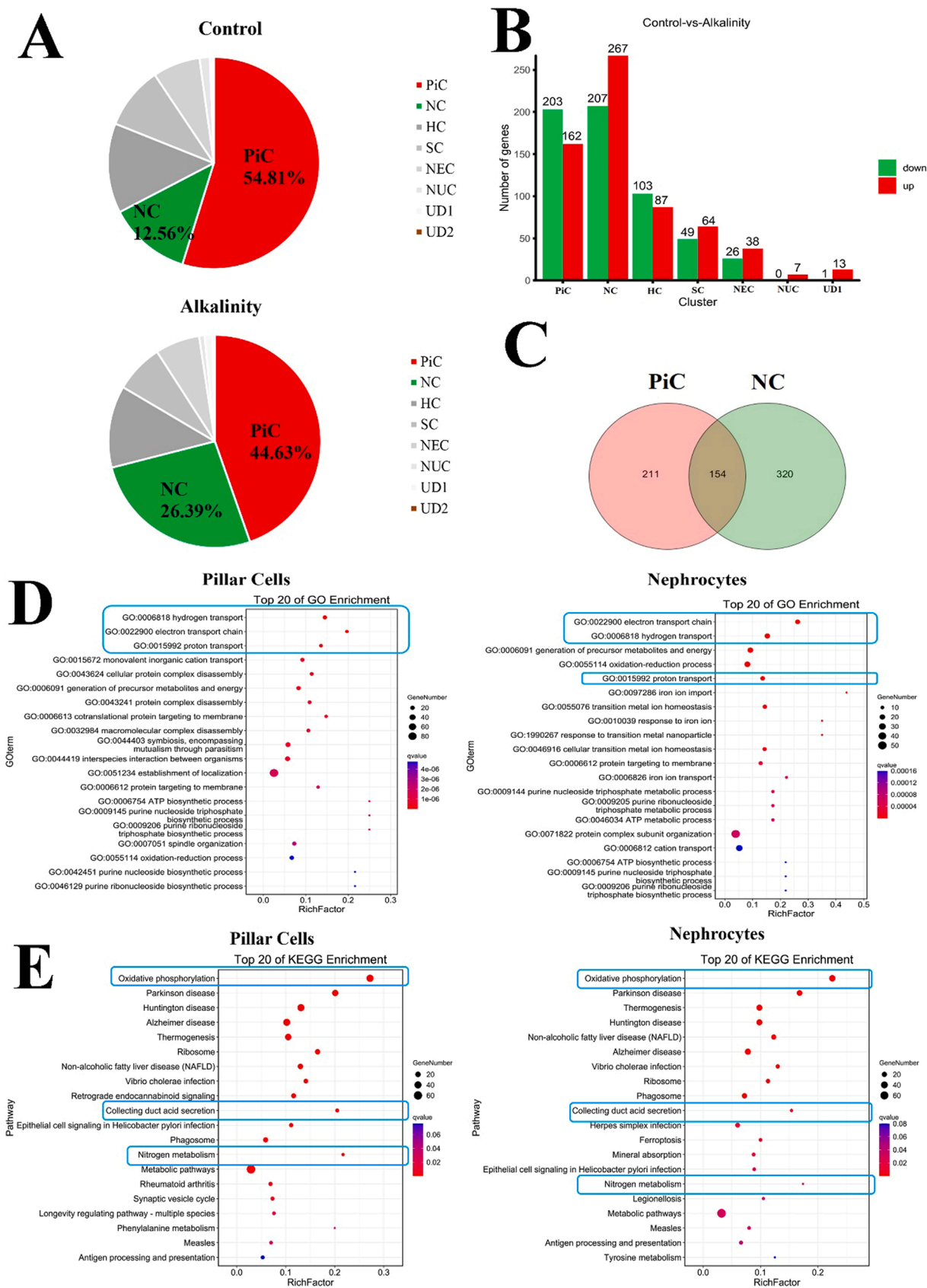


Fig. 4. Gene expression characteristics of pillar cells and nephrocytes after alkalinity stress. (A) Pie charts of 8 clusters identified from gill in control and alkalinity stress group by t-SNE analysis. PiC: Pillar cells, NC: Nephrocytes, HC: Hemocytes, SC: Septal cell, NEC: Nerve cell, NUC: Nurse cell, UD1: Undefined cell 1, UD2: Undefined cell 2. (B) Down- and up-regulated DEG numbers in each cell type of gills after alkalinity stress. (C) Venn diagram analysis of DEGs in pillar cells and nephrocytes after alkalinity stress. (D) GO enrichment of DEGs in pillar cells and nephrocytes. (E) KEGG enrichment of DEGs in pillar cells and nephrocytes.

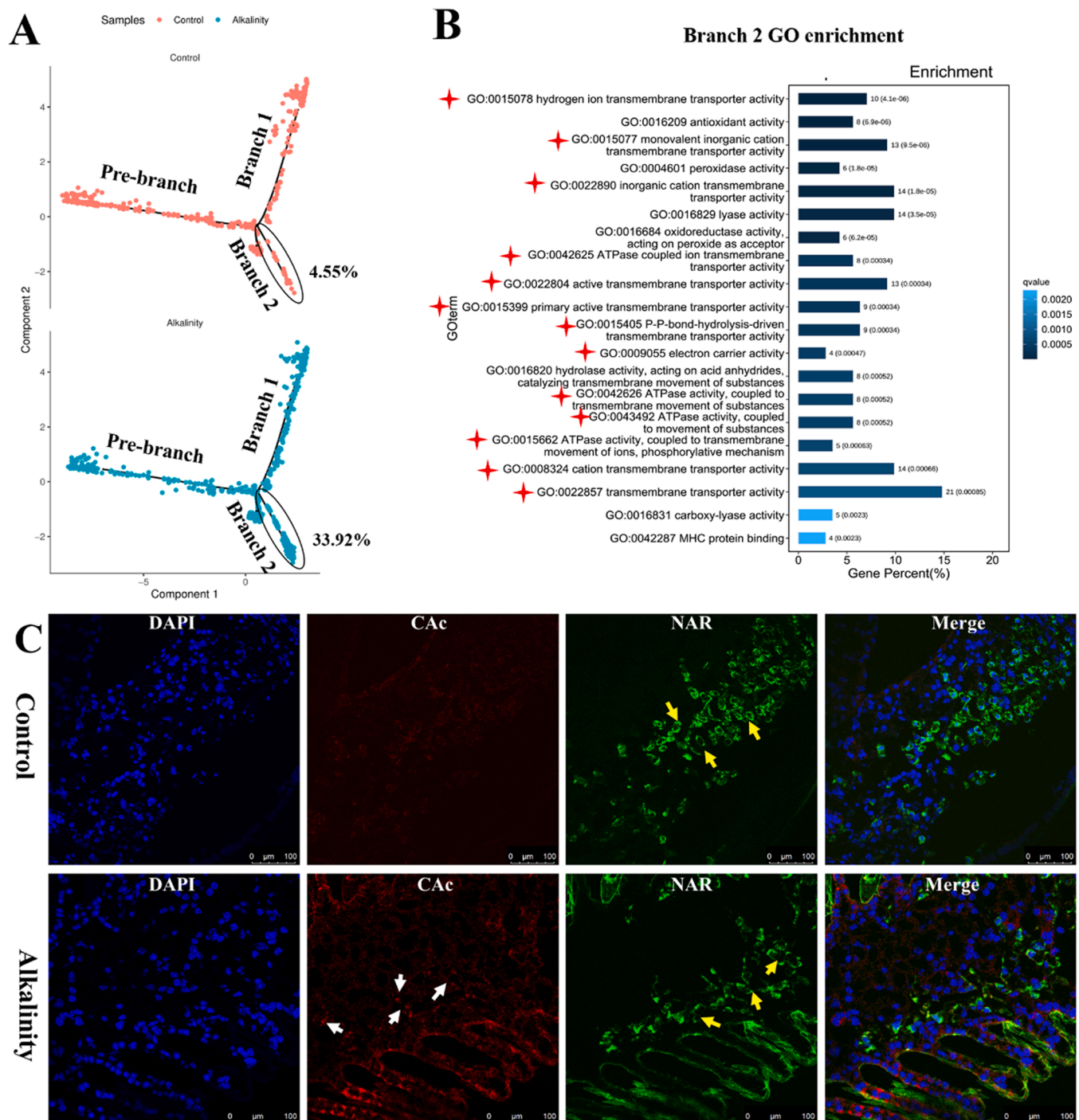


Fig. 5. Pseudotime analysis of nephrocytes. (A) Pseudotime analysis of DEGs in nephrocytes. (B) GO analysis of DEGs in Branch 2. (C) Changes of CAC (white arrow) and NAR (yellow arrow) in the nephrocytes of gill axis after alkalinity stress.

reprogrammed the differentiation trajectory of nephrocytes, and induced the production of branch 2 which was involved in ion transport, especially acid-base regulation. In view of the pillar cell cluster was mainly enriched in ion transport, the same function as branch 2 of nephrocyte cluster, which suggested that alkalinity stress may lead to the functional transformation of nephrocyte into pillar cell.

In order to verify this, the signal of CAC, the marker gene of pillar cells, was detected in the nephrocytes. The NAR-positive signal became more intensive and larger surrounded the vacuoles of nephrocytes after alkalinity stress compared with the control group. Moreover, the CAC-positive signal was barely expressed in the nephrocytes before

alkalinity stress, while significantly increased after alkalinity stress (Fig. 5C).

4. Discussion

Investigating the mechanisms of alkalinity adaption in the shrimp gill has been limited due to our lack of understanding of the cell types involved and the molecular basis of the process. In this study, scRNA-seq was successfully performed on shrimp gill for the first time. The comprehensive scRNA-seq data provide information on the cell populations and their stress-activated states in gills. Importantly, the scRNA-

seq could clearly distinguish 6 gill cell populations and their cellular localization, which are less well understood and for which marker genes were not previously available. Moreover, transcriptional changes in pillar cells and nephrocytes following exposure to high alkalinity were demonstrated.

In many aquatic animals, a rise in alkalinity caused acid-base disturbance, manifested in increased blood osmolality and pH. Yesaki and Iwama (1992) reported that chronic exposure to alkaline water (pH 10) induces a significant decrease in the hematocrit values of rainbow trout, accompanied by blood alkalosis. In *Colossoma macropomum*, the alkaline pH caused an increase in the mucus secretion, which affected osmoregulation (Aride et al., 2007). In *Magadi tilapia*, alkaline water caused the increase of Na^+ , HCO_3^- ion concentration in blood, which leads to the increase of osmolality (Wood et al., 2012). Wilkie and Wood (1996) reported that the increase of fish blood HCO_3^- concentration led to the increase of blood pH, which destroyed the acid-base balance in the body and leads to respiratory alkalosis. Therefore, the ability of aquatic animals to cope with alkalinity stress is mediated by acid-base regulation and osmoregulatory responses (Yao et al., 2015). In this study, hemolymph osmolality and pH of *E. carinicauda* were also increased in the first 48 h of alkalinity stress, which called crisis period. After 48 h, the hemolymph osmolality and pH gradually decreased and tended to stable, which called stabilization period (Partridge and Jenkins, 2002; Handeland et al., 2003). Ionoregulatory disturbance was directly connected to pH, and this may be owing to the inhibition of active ion uptake and increase of diffusive ion loss in the gills under pH stress conditions.

The gill epithelium is a heterogeneous assembly of epithelial cells that serve different structural and/or physiological functions (Evans et al., 2005). Diffusion of ions is proportional to surface area and inversely proportional to diffusion distance. To maximise oxygen uptake across the gills during metabolic challenges such as hypoxia, elevated temperature or exercise, functional and morphological changes occur in the gills to increase surface area and reduce diffusion distance (Zimmer and Perry, 2022). In this study, we also observed lengthened lateral flanges of pillar cells and enlarged vacuoles of nephrocytes after high alkalinity stress. It was similar to that observed in *Procambarus clarkii*, which is probably due to the blocked ion transport and increased osmotic pressure in the gill cell (Xiao et al., 2022). As for the molecular regulation mechanism of pillar cells, nephrocytes or other cells in response to alkalinity stress, further study was needed by scRNA-seq.

At present, only several cell types of gills such as pillar cell, septal cell and nephrocyte in crustacean have been observed by light and electron micrograph sections, other cell types have yet to be identified. In aquatic animals, Pan et al. (2022) presented the first transcriptomic analysis of cells from the zebrafish gills using scRNA-seq and defined 16 separate cell types. Our study is the first report to identify gill cell types of aquatic crustaceans at the single-cell level, and provides important information on their location and function. In this study, most ion-transport genes including VHAs, NCC, CA and Rh identified in the pillar cells in *E. carinicauda*, confirming the identity of this cell type. Meanwhile, the result of GO enrichment in pillar cells was mainly related to ion transport, especially related to hydron transport. From some previous studies, the pillar cells in gills of crustacean appear to be mainly involved in ion transport, given their intimate connection with both the external medium and the hemolymph (Faleiros et al., 2010; Thabet et al., 2017; Mcnamara et al., 2015). In the predicted models for Na^+ absorption across the gill lamellae of palaemonid shrimp *M. amazonicum*, the pillar and septal cells are structurally and functionally linked, that is Na^+ would enter pillar cells through apical ion exchangers and/or channels located in the extensive evaginations and it would pass through basal junctions into the septal cell from where it is actively exchanged for K^+ to the hemolymph via the NKA (Boudour-Boucheker et al., 2014). In a recent report, NKA distribution was described in gill septal cells in *Macrobrachium amazonicum* (Boudour-Boucheker et al., 2014). It thus appears that the septal cells also

play important role in ion transport. The GO terms enrichment of septal cells including “electron transport chain” confirmed it. Moreover, the hemocyte in gill was identified in this study. Shrimp, as invertebrates, have an open vasculature that allows circulating hemocytes to infiltrate the tissues, such as gills, where they are referred to as sessile hemocytes (Venediktova et al., 2013). So far, most studies of shrimp hemocyte have focused on circulating hemocytes, while little is known about the sessile hemocytes. The result in this study will provide more information on function and molecular markers of gill sessile hemocytes. It is worth noting that the nephrocytes, nerve cell and nurse cell with little information were also identified by scRNA-seq and located by FISH technology. Especially the nephrocyte, localized in the central gill axis, have a voluminous vacuole, surrounded by satellite vacuoles, and a lateral nucleus, which was undistinguished from those already described in the gills of other crustaceans (Petriella and Sousa, 2005). The previous study only found the existence of nephrocyte, but the knowledge of the function of this cell type remains limited. Only Taylor and Taylor (1992) suggested they function as a hemolymph molecular filter, resulting in the accumulation of foreign substances in the central vacuole. In addition, except for pillar cells and septal cells, the nerve cells and nurse cells were also found in the gill lamella of *E. carinicauda*, and the results of scRNA will provide some potential gene markers to further study their function.

In this study, scRNA-seq revealed significant variation of pillar cell and nephrocyte after alkalinity stress, such as decreased cell proportion and down-regulated acid-base regulation gene expressions in pillar cells and increased cell proportion and up-regulated acid-base regulation gene expressions in nephrocytes. Based on our KEGG analysis, three pathways involved in acid-base regulation were enriched by DEGs: “Oxidative phosphorylation”, “Collecting duct acid secretion” and “Nitrogen metabolism”. In these pathways, acid-base transport proteins, especially VHAs and CAs, were found to be involved in, and all of these gene expressions were decreased in the pillar cells and increased under alkalinity stress. VHAs are enzymes that operate through a rotary mechanism resulting in unidirectional H^+ transport across the membrane (Capasso et al., 2021; Fehsenfeld and Weihrauch, 2017; Perry et al., 2003). In some crustaceans, VHAs located apically in the gill pillar cell flanges may complement the intralamellar septal cell NKA in energizing osmoregulatory NaCl uptake, and also participate in H^+ export (Boudour-Boucheker et al., 2014; Li et al., 2021). In elasmobranch species, basolateral H^+ -ATPase activity enables H^+ excretion into the blood compartment, thereby facilitating net H^+ uptake (i.e. base excretion) as a mechanism to regulate blood alkalosis (Piermarini et al., 2002; Tresguerras et al., 2005). In the dogfish shark, a novel mechanism was described whereby H^+ -ATPase in a sub-population of gill ionocytes is redistributed from cytoplasm to the basolateral membrane during systemic alkalosis owing to the stimulation of soluble adenylyl cyclase by elevated levels (Tresguerras et al., 2010, 2019). Therefore, down-regulation suggests that acid-base regulated function related to VHAs is impaired in pillar cells after alkalinity stress. CA is a type of zinc metalloenzymes that catalyze the interconversion of CO_2 and HCO_3^- . CAC exists in the cytoplasm and provides counter ions (H^+ and HCO_3^-) for the Na^+/H^+ and $\text{Cl}^-/\text{HCO}_3^-$ exchanger, driving ion uptake in gills. Glycosyl-phosphatidylinositol-linked carbonic anhydrase (CAG) is mainly located on the cell membrane and has been suggested to facilitate the mobilization of hemolymph HCO_3^- to molecular CO_2 and assists with CO_2 excretion via the gills. Previous research has shown that the expression of CAs were strongly induced in crustaceans by changes in water pH or alkalinity (Ge et al., 2019; Ali et al., 2015; Liu et al., 2015). To compensate for the alkalosis caused by high alkalinity, *Gymnocypris przewalskii* down-regulated branchial CAC in the epithelium and would thereby be expected to produce less H^+ for the H^+ -ATPase and NHEs which are the main acid excretion pathways. On the other hand, down-regulation of branchial CAC would also decrease the concentration of HCO_3^- and H^+ in the cell, which are used to exchange for Cl^- and Na^+ into the cell, respectively (Yao et al., 2016). Similarly, it was

speculated that *E. carinicauda* decreased VHAs and CAs expressions in pillar cells to compensate for a respiratory alkalosis. On the other hand, their increased expressions in nephrocytes will promote HCO_3^- excretion and H^+ uptake in order to maintain acid-base balance in response to alkalinity stress.

Based on the pseudotime analysis of nephrocyte, the branch 2 consists of more than half of the nephrocytes in alkalinity stress group, and most of its DEGs were involved in the process of ion transport, especially acid-base regulation. In view of the pillar cell cluster was mainly enriched in ion transport, the same function as branch 2 of nephrocyte cluster, which suggested that alkalinity stress may lead to the functional transformation of nephrocyte into pillar cell. This is the first time to report this new viewpoint. The traditional view was that the enzymes or proteins associated with ion transport were mainly located in the pillar cell of crustacean gill, and take it for granted that the pillar cell played the most important role in response to environment stress including salinity, pH or other stress (Antunes et al., 2017; Boudour-Boucheker et al., 2013; Pinto et al., 2016). However, in this study, no matter the cell morphology or differentiation characteristics, suggested the more important role of the nephrocyte than pillar cell in response to alkalinity stress. In addition, the increased CAC-positive signal in the nephrocytes after alkalinity stress provides the compelling evidence that nephrocytes replaced pillar cell to play positive role in ion transport to maintain shrimp acid-base balance in response to alkalinity stress. Beyond that, although the ion transport function of the pillar cell is temporarily restrained after alkalinity stress, its distribution obviously changed, became more symmetrically, accompanied with obviously decrease of hemolymph space. In some crustaceans, two pillar cells may span the entire hemolymph space across the gill lamella. They sustained the intralamellar septum gill lamellae and limited deformation of the hemolymph space during changes in hydrostatic pressure (Freire et al., 2008). Therefore, we speculated that pillar cell may exert the structural function exclusively, and the nephrocyte is responsible for acid-base regulation after alkalinity stress.

5. Conclusions

This study presents the first report of the utilization of scRNA-Seq analysis for the study of the cellular and molecular mechanism of shrimp gills adaption to alkalinity stress. At first, alkalinity stress caused acid-base disturbance of hemolymph and structural alterations of pillar cells and nephrocytes in gills for *E. carinicauda* obviously. Then, scRNA-seq results of gills showed that the acid-base regulation of the pillar cells was inhibited after alkalinity stress and the function was replaced by the nephrocytes. Future studies will verify more marker genes of the nephrocytes and determine how the genes contribute to acid-base regulation. Results of this study provide further understanding of the crustacean's response to alkalinity stress, and further insight into its adaption.

CRediT authorship contribution statement

Qianqian Ge: Investigation, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Jitao Li:** Conceptualization, Methodology, Supervision, Resources, Funding acquisition, Project administration, Writing – review & editing. **Jiajia Wang:** Validation, Methodology, Data curation, Writing – review & editing. **Jian Li:** Conceptualization, Methodology, Supervision, Resources, Funding acquisition, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2023.114902.

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